Лозовий Олег КН-415
Варіант 12

2. Point out the wrong statement.
b) Soft computing represents a real paradigm shift in the way in which systems are deployed
Because soft computing represents a shift in how systems handle uncertainty, imprecision, and approximation, rather than how they are deployed, so it focuses on improving system intelligence rather than deployment strategies.

12. Point out the correct statement.
c) Data centers have been sited to optimize the overall system latency
Because data centers are often strategically located to reduce latency and provide faster access to data and services for users and applications.

22. Point out the correct statement.
c) Cloud computing presents new opportunities to users and developers
Because cloud computing offers new opportunities by providing on-demand access to scalable resources, enabling users and developers to innovate and improve flexibility without the need for extensive on-premises infrastructure.

32. What is a public cloud?
c) A multi-tenant cloud environment accessed over the internet
Because a public cloud refers to a cloud computing environment where resources are owned and operated by 3th-party provider and shared by multiple tenants, accessible over the internet.

42. Point out the correct statement.
a) Cloud computing is a natural extension of many of the design principles, protocols, plumbing, and systems
Because cloud computing builds upon and extends existing design principle and protocols to provide scalable on-demand computing resources over the internet.

52. Point out the correct statement.
d) All of the mentioned
Because each statement accurately describes key aspects of cloud computing and software platforms. Cloud architecture enables scalable services, protocols manage communication, and platforms help build complex applications, all of which are integral to modern computing environments.

62. Point out the correct statement.
a) Virtual machines are containers that are assigned specific resources
Because virtual machines are isolated environments that run their own operating systems and are allocated specific resources from the host system.

72. Point out the wrong statement.
d) None of the mentioned
Because all the statements are essentially accurate. Statement a is true as cloud computing allows for flexible deployment of applications. Statement b is not entirely incorrect, as while the location may influence how an application is optimized or deployed, it doesn't fundamentally change the application's written logic. Finally, c correctly describes how applications run on traditional systems like desktops or servers.

82. _____ empowers IoT by bringing together everyday objects.
b) Connectivity
Because connectivity empowers IoT by enabling everyday objects to communicate, share data, and work together through networks.

92. Which is an open standard?
b) MQTT
Because it is open and standardized by OASIS and ISO for use in IoT.

102. What is the microcontroller used in Arduino UNO?
a) ATmega328p
As documentation said: because it is a low-power, high-performance 8-bit AVR microcontroller with sufficient memory and peripherals to support a wide range of embedded applications.

112. Fog computing can be perceived in _____ and _____
a) Big data and Cloud systems
Because it acts as a bridge by processing data closer to the edge before sending it to centralized cloud systems, thus optimizing big data analytics and reducing latency.

122. True or False? The purpose of the IBM IoT System is to create complexity in IoT system implementation and management.
False, because main purpose is to simplify implementation and management.

132. Match the functions with the IoT device.

adds intelligence to things	– controller
monitors physical properties of the environment	– sensor
improves IoT network performance	– none of these
performs an action	– actuator

This matching is correct because a controller processes data and makes decisions, adding intelligence to IoT systems, while a sensor detects and monitors environmental changes. An actuator performs actions based on controller decisions, and network performance is typically enhanced by components like gateways, not directly by any of these devices.